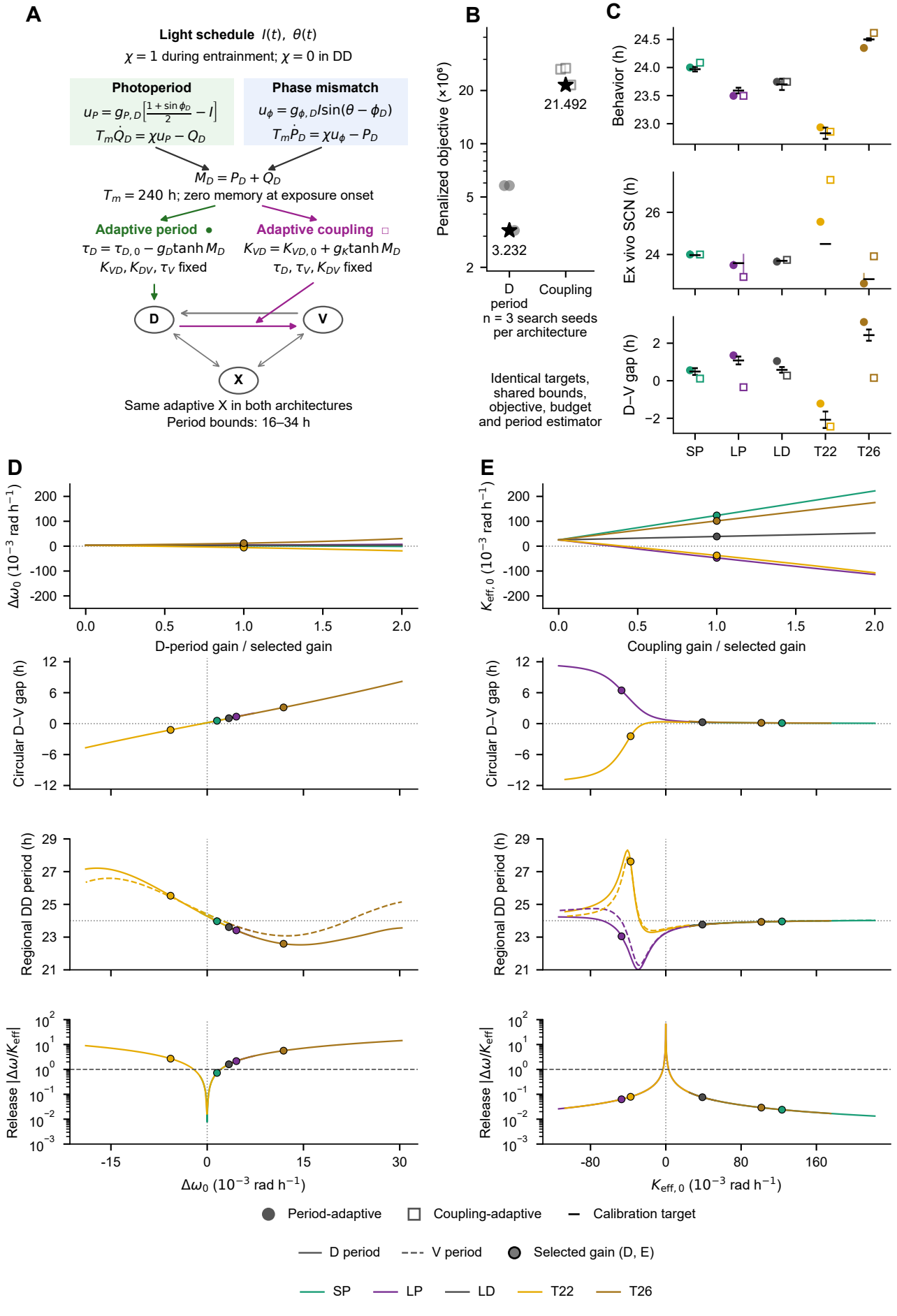

A
Light schedule $I(t)$, $\theta(t)$
$\chi = 1$ during entrainment; $\chi = 0$ in DD
Photoperiod
$u_P = g_{P,D}\left[\frac{1+\sin\phi_D}{2} - I\right]$
$T_m\dot{Q}_D = \chi u_P - Q_D$
Phase mismatch
$u_\phi = g_{\phi,D} I \sin(\theta - \phi_D)$
$T_m\dot{P}_D = \chi u_\phi - P_D$
$M_D = P_D + Q_D$
$T_m = 240$ h; zero memory at exposure onset
Adaptive period
$\tau_D = \tau_{D,0} - g_D \tanh M_D$
K_{VD}, K_{DV}, τ_V fixed
Adaptive coupling
$K_{VD} = K_{VD,0} + g_K \tanh M_D$
τ_D, τ_V, K_{DV} fixed
D
V
X
Same adaptive X in both architectures
Period bounds: 16–34 h
B
Penalized objective (×10⁶)
21.492
3.232
D period
Coupling
n = 3 search seeds per architecture
Identical targets, shared bounds, objective, budget and period estimator
C
Behavior (h)
Ex vivo SCN (h)
D–V gap (h)
SP
LP
LD
T22
T26
D
$\Delta\omega_0$ (10^{-3} rad h^{-1})
D-period gain / selected gain
Circular D–V gap (h)
Regional DD period (h)
Release $|\Delta\omega/K_{\text{eff}}|$
$\Delta\omega_0$ (10^{-3} rad h^{-1})
E
$K_{\text{eff},0}$ (10^{-3} rad h^{-1})
Coupling gain / selected gain
Circular D–V gap (h)
Regional DD period (h)
Release $|\Delta\omega/K_{\text{eff}}|$
$K_{\text{eff},0}$ (10^{-3} rad h^{-1})
Period-adaptive
Coupling-adaptive
Calibration target
D period
V period
Selected gain (D, E)
SP
LP
LD
T22
T26